

Today's Agenda

- **Color image processing**
- **Image Compression**

Color Models

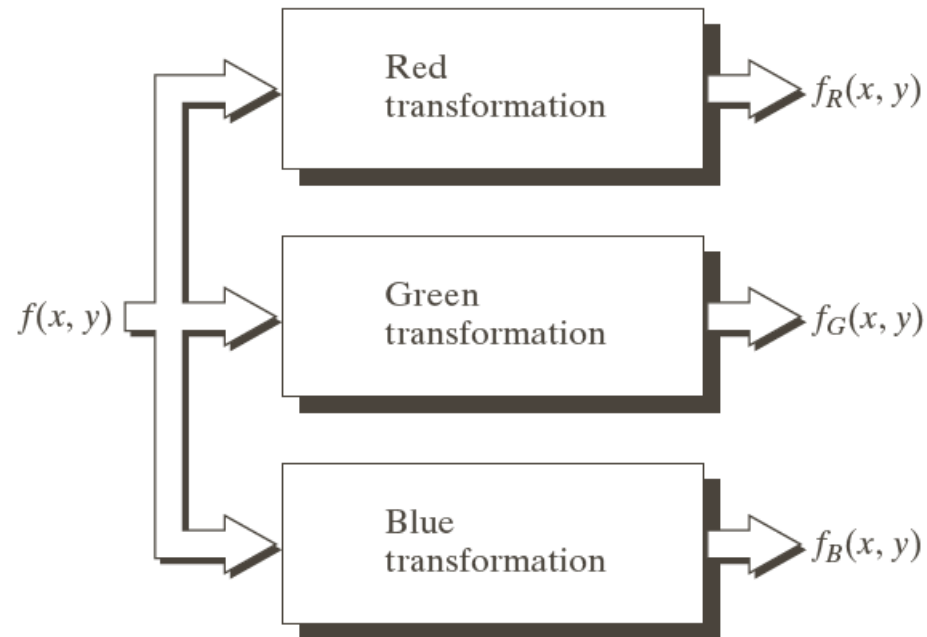
Color model (space/system): a coordinate system or a subspace to represent the colors

- RGB model: monitors and cameras
- CMY (Cyan, magenta, and yellow): printing
- HSI (Hue, saturation, and intensity): separate color and gray level information

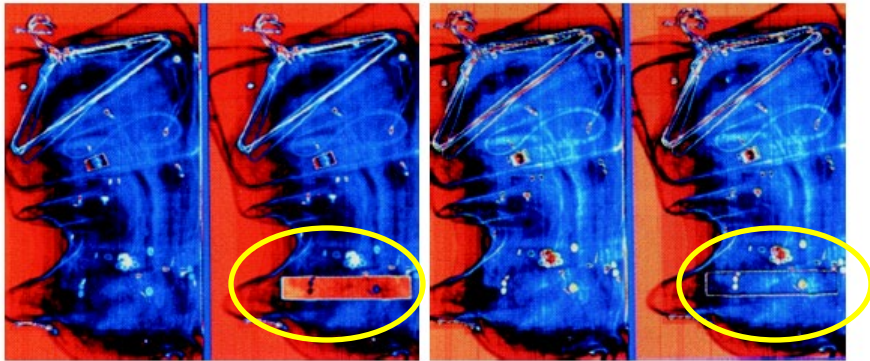
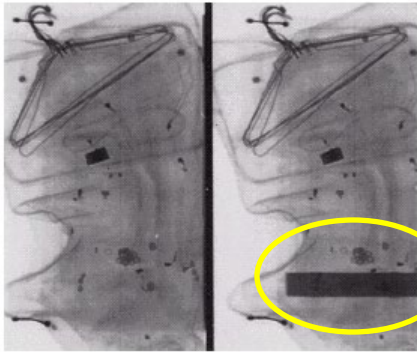
Intensity to Color Transformation

FIGURE 6.23

Functional block diagram for pseudocolor image process: f_R , f_G , and f_B fed into the corresponding red, green, and blue inputs of RGB color monitor.

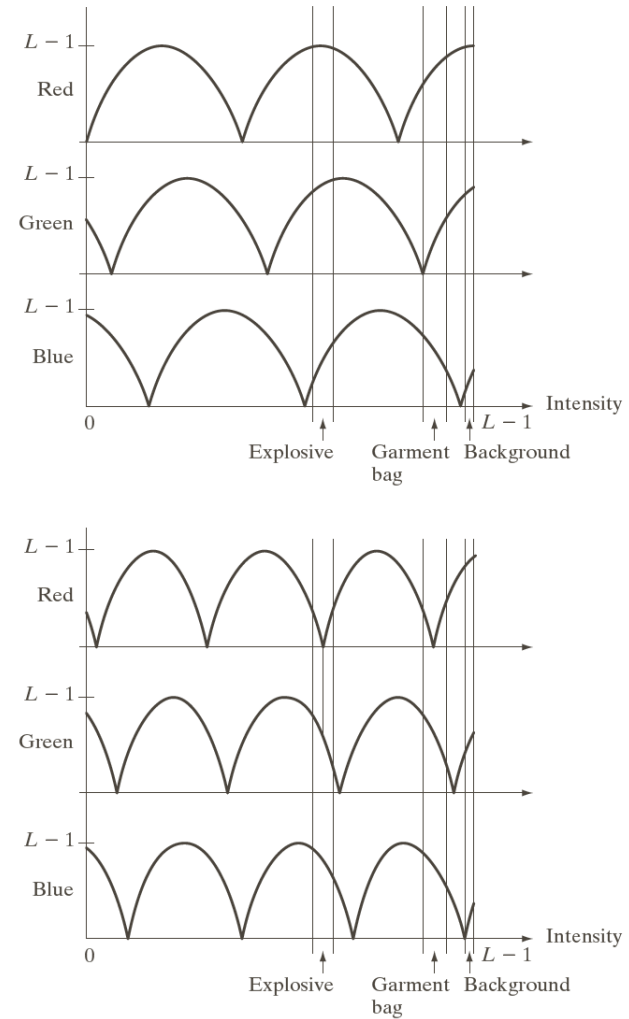


Example



a
b c

FIGURE 6.24 Pseudocolor enhancement by using the gray-level to color transformations in Fig. 6.25. (Original image courtesy of Dr. Mike Hurwitz, Westinghouse.)



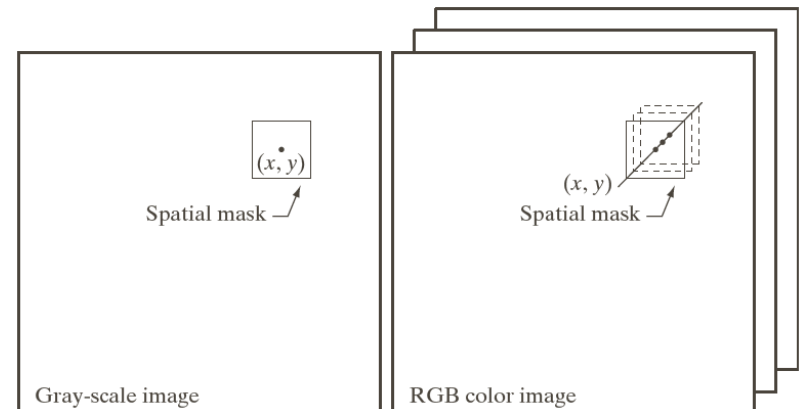
a
b

FIGURE 6.25 Transformation functions used to obtain the images in Fig. 6.24.

Full-color Image Processing

Pixel in color image $\mathbf{p}(x, y) = \begin{bmatrix} p_r(x, y) \\ p_g(x, y) \\ p_b(x, y) \end{bmatrix}$

- **Process each component/channel individually, then generate the composite image**
- **Work on each pixel individually**



Color Transformation

For a color image with n components

input values for all components

$$s_i = T_i(r_1, r_2, \dots, r_n), \quad i = 1, 2, \dots, n$$

Output value for i^{th} component

Transformation functions

- Modify intensity
- Color complement (“negative” color image)
- Color slicing
- Tonal correction
- Color balancing
- Histogram processing

Examples of Color Image Transformation

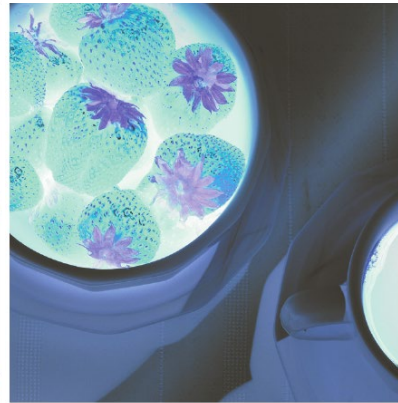


Original image



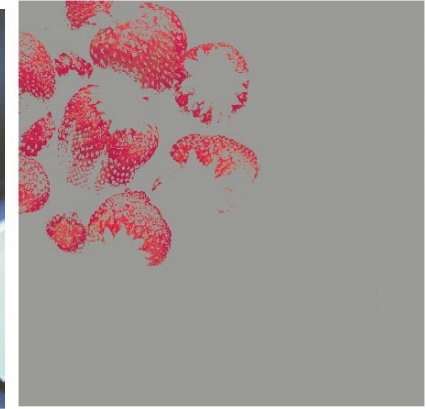
Intensity
modification

HSI



Complement
color

RGB



Color slicing

RGB

Tonal Correction

Correct the tonal range (distribution of color intensities)

- Recall the intensity transformation in the gray level images
- For RGB model, each component has the same transformation function
- For HSI model, the transformation is applied on the intensity component only

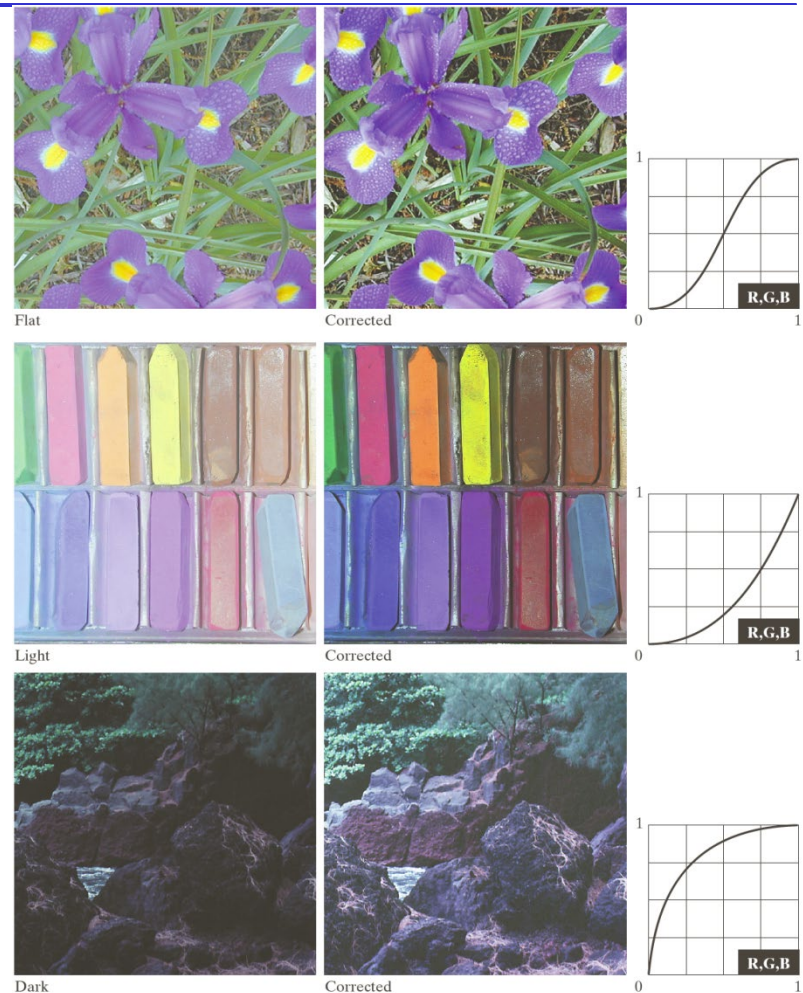
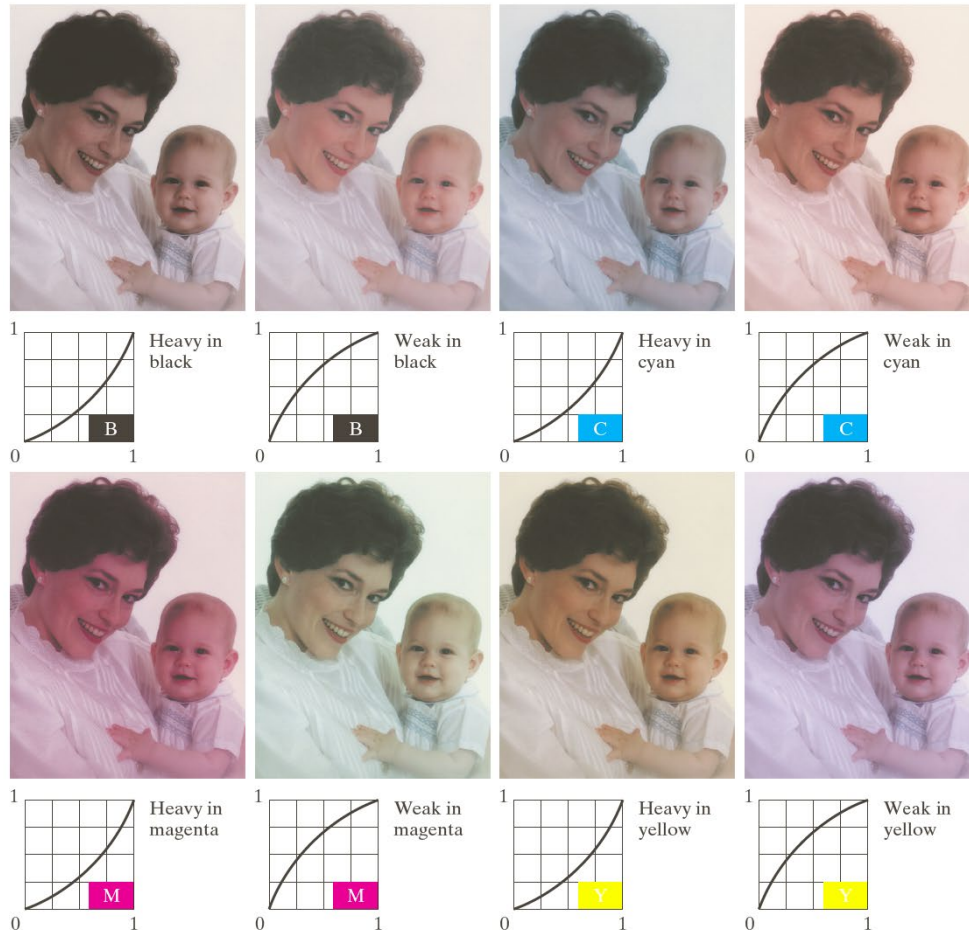


FIGURE 6.35 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not always alter the image hues significantly.

Color Balancing

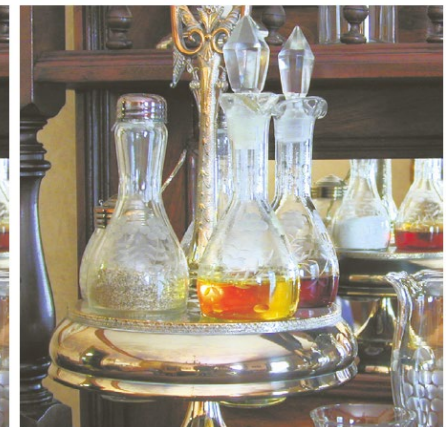
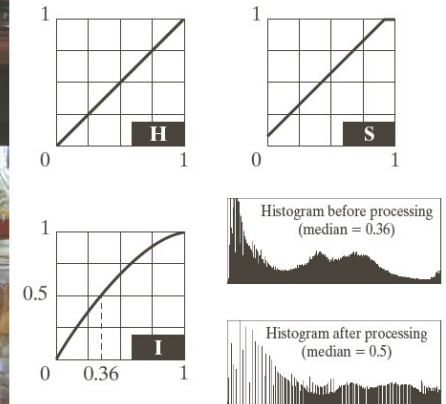
Correct color unbalance by analyzing a known color in image



Histogram Processing

Step 1: Histogram equalization

Step 2: Saturation adjustment



a	b
c	d

FIGURE 6.37
Histogram equalization (followed by saturation adjustment) in the HSI color space.

Reading Assignment

- **Reading Chapter 6.6, 6.7, 6.8**
- **Read Chapter 7 (Wavelets and Multiresolution Processing)**

Image Compression

Why?

Standard definition (SD) television movie (raw data)

$$30 \frac{\text{frames}}{\text{sec}} \times (720 \times 480) \frac{\text{pixels}}{\text{frame}} \times 3 \frac{\text{bytes}}{\text{pixel}} = 31,104,000 \text{ bytes/sec}$$

A two-hour movie

$$31,104,000 \frac{\text{bytes}}{\text{sec}} \times (60^2) \frac{\text{sec}}{\text{hour}} \times 2 \text{ hours} \approx 224 \text{ GB}$$

Need 27 8.5GB dual-layer DVDs!

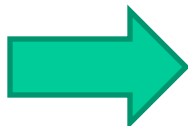
High-definition (HD) television 1920x1080x24 bits/image!

Image Compression (Cont'd)

Standard definition (SD) television movie (raw data)

$$30 \frac{\text{frames}}{\text{sec}} \times (720 \times 480) \frac{\text{pixels}}{\text{frame}} \times \frac{24\text{bits}}{\text{pixel}} = 248,832,000 \text{bit/sec} > 200 \text{Mbit/sec}$$

WAN modems	Ethernet LAN	WiFi WLAN	Mobile data
1972: Acoustic coupler 300 baud 1977: 1200 baud Vadic and Bell 212A 1986: ISDN introduced with two 64 kbit/s channels (160 kbit/s gross bit rate) 1990: v.32bis modems: 2400 / 4800 / 9600 / 19200 bit/s 1994: v.34 modems with 28.8 kbit/s 1995: v.90 modems with 56 kbit/s downstreams, 33.6 kbit/s upstreams 1999: v.92 modems with 56 kbit/s downstreams, 48 kbit/s upstreams 1998: ADSL up to 8 Mbit/s, 2003: ADSL2 up to 12 Mbit/s 2005: ADSL2+ up to 24 Mbit/s	1972: IEEE 802.3 Ethernet 2.94 Mbit/s 1985: 10b2 10 Mbit/s coax thinwire 1990: 10bT 10 Mbit/s 1995: 100bT 100 Mbit/s (125 Mbit/s gross bit rate) 1999: 1000bT (Gigabit) 1 Gbit/s (1.25 Gbit/s gross bit rate) 2003: 10GBASE 10 Gbit/s http://en.wikipedia.org/wiki/Bit_rate	WiFi WLANs 1997: 802.11 2 Mbit/s 1999: 802.11b 11 Mbit/s 1999: 802.11a 54 Mbit/s (72 Mbit/s gross bit rate) 2003: 802.11g 54 Mbit/s (72 Mbit/s gross bit rate) 2005: 802.11g (proprietary) 108 Mbit/s 2007: 802.11n 600 Mbit/s	1G: 1981: NMT 1200 bit/s 2G: 1991: GSM CSD and D-AMPS 14.4 kbit/s 2003: GSM EDGE 296 kbit/s down, 118.4 kbit/s up 3G: 2001: UMTS-FDD (WCDMA) 384 kbit/s 2007: UMTS HSDPA 14.4 Mbit/s 2008: UMTS HSPA 14.4 Mbit/s down, 5.76 Mbit/s up 2009: HSPA+ (Without MIMO) 28 Mbit/s downstreams (56 Mbit/s with 2x2 MIMO), 22 Mbit/s upstreams 2010: CDMA2000 EV-DO Rev. B 14.7 Mbit/s downstreams - Pre-4G: 2007: Mobile WiMAX (IEEE 802.16e) 144 Mbit/s down, 35 Mbit/s up. 2009: LTE 100 Mbit/s downstreams (360 Mbit/s with MIMO 2x2), 50 Mbit/s upstreams See also Comparison of mobile phone standards



3.5Mbits/sec with MPEG-2 compression

Fundamentals

Data represent information – different ways

Representations that contain irrelevant or repeated information → contain redundant data

Two representations of the same information: b and b' bits, then the relative data redundancy

$$R = 1 - \frac{1}{C}, \quad \text{where}$$

$$C = \frac{b}{b'} \text{ is called the compression ratio}$$

In digital image processing, b is the # bits for the 2D array representation and b' is the compressed representation

Three Types of Data Redundancies

Coding redundancy

- Code/code book is a system to represent information
- Code length is the number of symbols in each code word
- Do we really need 8 bits to represent a gray-level pixel?

Spatial and temporal redundancy

- Neighboring (spatially or temporally) pixels usually have similar intensities!
- Do we need to represent every pixel?

Irrelevant information

- Some image information can be ignored.

Coding Redundancy

Histogram

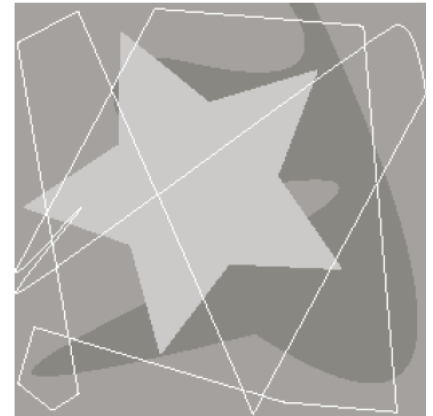
$$p_r(r_k) = \frac{n_k}{MN}, \quad k = 0, 1, 2, \dots, L-1$$

Average # bits required to represent each pixel

$$L_{\text{avg}} = \sum_{k=0}^{L-1} l(r_k) p_r(r_k)$$

Number of bits representing each intensity level

Total bits MNL_{avg}



r_k	$p_r(r_k)$	Code 1	$l_1(r_k)$	Code 2	$l_2(r_k)$
$r_{87} = 87$	0.25	01010111	8	01	2
$r_{128} = 128$	0.47	10000000	8	1	1
$r_{186} = 186$	0.25	11000100	8	000	3
$r_{255} = 255$	0.03	11111111	8	001	3
r_k for $k \neq 87, 128, 186, 255$	0	—	8	—	0

TABLE 8.1
Example of
variable-length
coding.

Fixed length

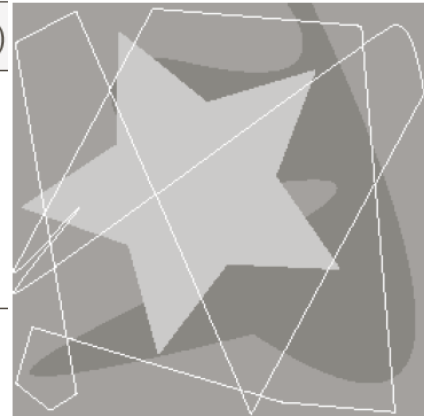
Variable length

$$L_{\text{avg}} = 2 * 0.25 + 0.47 * 1 + 0.25 * 3 + 0.03 * 3 = 1.8$$

Coding Redundancy

TABLE 8.1
Example of
variable-length
coding.

r_k	$p_r(r_k)$	Code 1	$l_I(r_k)$	Code 2	$l_2(r_k)$
$r_{87} = 87$	0.25	01010111	8	01	2
$r_{128} = 128$	0.47	10000000	8	1	1
$r_{186} = 186$	0.25	11000100	8	000	3
$r_{255} = 255$	0.03	11111111	8	001	3
r_k for $k \neq 87, 128, 186, 255$	0	—	8	—	0



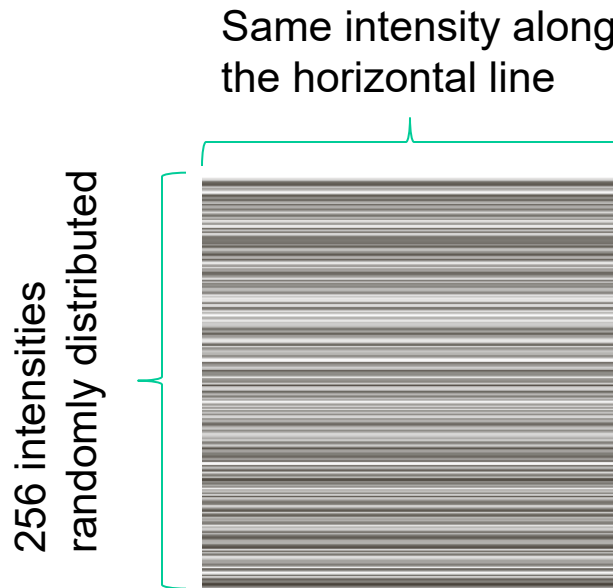
C and R with variable length coding?

$$C = \frac{8}{1.81} = 4.42$$

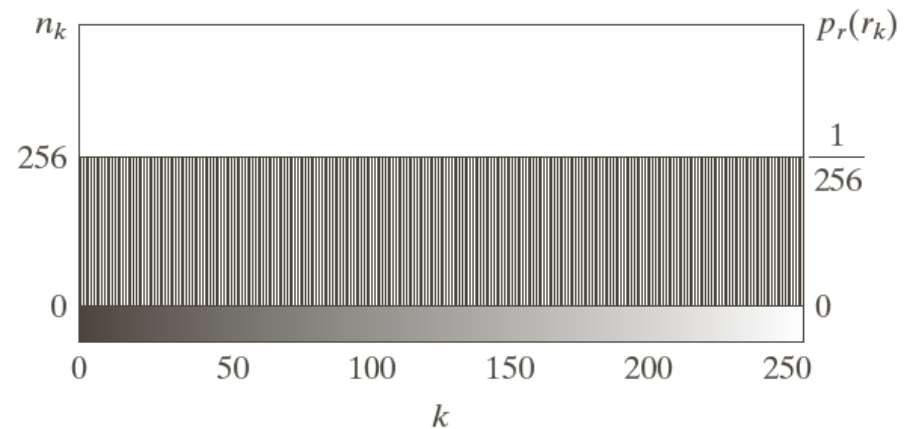
$$R = 1 - \frac{1}{C} = 0.77$$

Spatial and Temporal Redundancy

FIGURE 8.2 The intensity histogram of the image in Fig. 8.1(b).



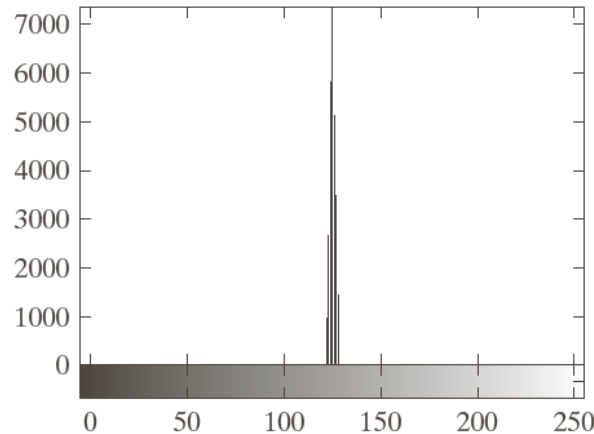
The histogram is uniform.



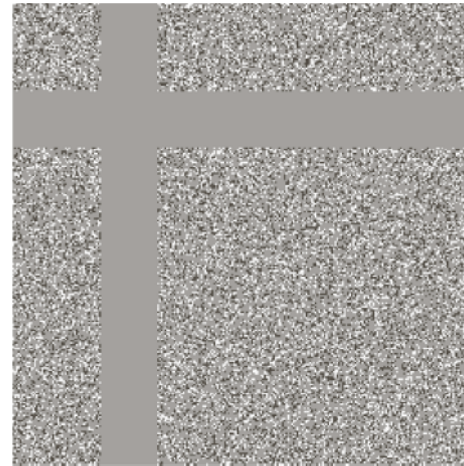
Compression by mapping:

- Run-length coding:
 - one word representing the intensity, and one word representing the length
 - C and R?
- Difference between two neighboring pixels

Irrelevant Information



Represented by a single byte



a b

FIGURE 8.3
(a) Histogram of the image in Fig. 8.1(c) and (b) a histogram equalized version of the image.

More details

**Quantization – loss of quantitative information:
irreversible operation**

Measuring Image Information

Minimum amount of data without losing information?

A random event E with probability $P(E)$ contains information

$$I = \log \frac{1}{P(E)} = -\log P(E)$$

Entropy (average information per image intensity)

$$H = -\sum_{k=0}^{L-1} p_r(r_k) \log_2 p_r(r_k)$$

Shannon's first theorem (noiseless coding theorem)

$$L_{\text{avg}} \geq H$$

and the low-bound H can be achieved by a coding method

Fidelity Criteria – Quantify the Loss

objective fidelity criteria

Root mean square error

$$e_{\text{rms}} = \left[\frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]^2 \right]^{\frac{1}{2}}$$

Mean-square signal to noise ratio

$$SNR_{\text{ms}} = \frac{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \hat{f}(x, y)^2}{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]^2}$$

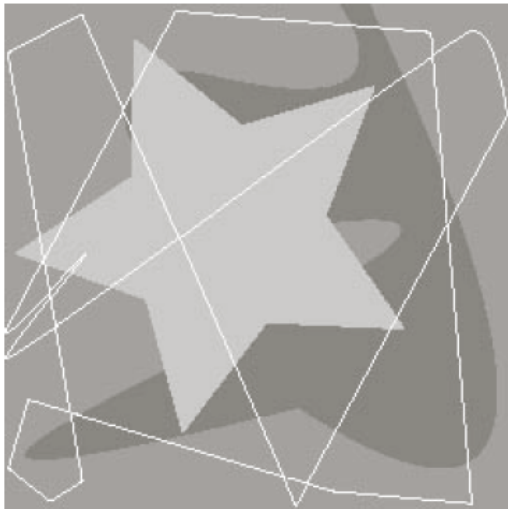
TABLE 8.2
 Rating scale of
 the Television
 Allocations Study
 Organization.
 (Freundtall and
 Behrend.)

Subjective Fidelity Criteria

Value	Rating	Description
1	Excellent	An image of extremely high quality, as good as you could desire.
2	Fine	An image of high quality, providing enjoyable viewing. Interference is not objectionable.
3	Passable	An image of acceptable quality. Interference is not objectionable.
4	Marginal	An image of poor quality; you wish you could improve it. Interference is somewhat objectionable.
5	Inferior	A very poor image, but you could watch it. Objectionable interference is definitely present.
6	Unusable	An image so bad that you could not watch it.

Inconsistency between Objective and Subjective Fidelity Criteria

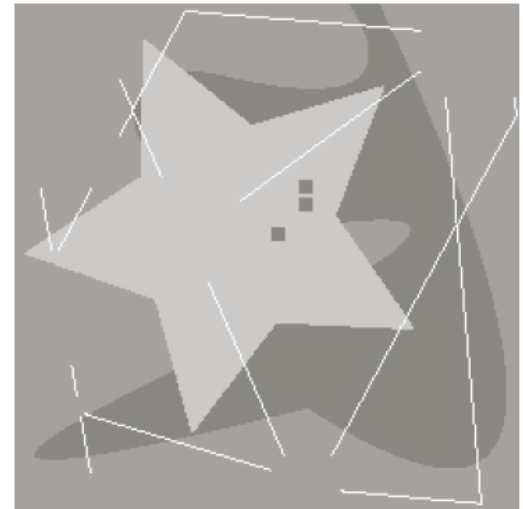
Objective: rms = 5.17



rms = 15.67



rms = 14.17



Subjective: Excellent

Passable

unusable

a b c

FIGURE 8.4 Three approximations of the image in Fig. 8.1(a).